

Data Model Documentation

1. Data Modeling Approach

The project follows a **star-schema-inspired model**:

- **Fact Table:** Sales Transactions
- **Dimension Tables:** Products, Regions

This approach simplifies analytical queries and improves performance for reporting tools such as Power BI.

2. Source Data Model

2.1 Sales Transactions (Fact)

Purpose

Captures individual sales events occurring across retail stores.

Column Name	Data Type	Description
transaction_id	STRING	Unique transaction identifier
transaction_date	DATE	Date of transaction
store_id	STRING	Store where sale occurred
product_id	STRING	Product sold
quantity	INT	Number of units sold
unit_price	DECIMAL	Price per unit
discount	DECIMAL	Discount applied
total_amount	DECIMAL	Final transaction amount
currency	STRING	Transaction currency
last_updated_ts	TIMESTAMP	Incremental tracking

2.2 Products (Dimension)

Purpose

Provides product-level reference information.

Column Name	Data Type	Description
product_id	STRING	Unique product identifier
product_name	STRING	Product name

Column Name	Data Type	Description
category	STRING	Product category
brand	STRING	Brand name
unit_cost	DECIMAL	Cost price
active_flag	BOOLEAN	Product status
last_updated_ts	TIMESTAMP	Change tracking

2.3 Stores (Dimension)

Purpose

Defines store locations and regional hierarchy.

Column Name	Data Type	Description
store_id	STRING	Unique store identifier
store_name	STRING	Store name
city	STRING	City
region	STRING	Sales region
country	STRING	Country
active_flag	BOOLEAN	Store status
last_updated_ts	TIMESTAMP	Change tracking

3. Layer-wise Data Model

3.1 Bronze Layer Model

- Stores raw data with original schema
- Adds ingestion metadata:
 - ingestion_timestamp
 - ingestion_date
 - source_system
- No transformations applied

3.2 Silver Layer Model

- Deduplicated using natural keys

- Validated against business rules
- Referential integrity enforced
- Invalid records stored separately

Silver Tables

- silver_sales_transactions
- silver_products
- silver_stores
- silver_sales_quarantine

3.3 Gold Layer Model

Gold tables represent aggregated business metrics and are optimized for analytics.

Gold Tables

Table Name	Description
daily_sales_summary	Daily revenue and volume metrics
monthly_revenue_region	Monthly revenue by region
product_performance	Product-level KPIs

4. Data Quality Handling

- Duplicate records removed in Silver
- Invalid references quarantined
- Missing attributes standardized
- Business rule violations corrected

5. Analytical Readiness

The data model supports:

- Time-series analysis
- Product and category analysis
- Regional performance analysis
- Executive reporting